# NECESSARY CONDITIONS FOR MULTIPLICITY FIVE AMONG BINOMIAL COEFFICIENTS

ARGUS OPEN RESEARCH PROJECT

Abstract. For an integer $a > 1$, let $P(a) = \{(n,k) \in \mathbb{Z}^2 : 1 \le k \le n/2,\ \binom{n}{k} = a\}$ denote the set of occurrences of $a$ in the left half of Pascal's triangle, let $m(a) = |P(a)|$ be its left-half multiplicity, and let $K(a) = \max\{k : (n,k) \in P(a)\}$ be the largest lower index among its representations. Erdős (Problem 849) and Singmaster asked how many times an integer can appear as a binomial coefficient; no integer with $m(a) \ge 5$ is known. Combining complete classical Diophantine classifications for low-index binomial collisions with numerator-window monotonicity and prime-support capacity bounds, we prove four unconditional global necessary conditions on any integer $a$ with $m(a) = 5$: (1) $K(a) \ge 8$; (2) $\omega(a) \ge 7$, where $\omega(a)$ denotes the number of distinct prime divisors of $a$; (3) at $K(a) = 8$, pairwise window-overlap reductions eliminate eighteen of the twenty candidate four-column systems, leaving only $\{3,5,7,8\}$ and $\{5,6,7,8\}$; and (4) every integer solution on the surviving $\{5,6,7,8\}$ branch satisfies $\omega(a) \ge 14$.

## 1. Introduction

Let $a > 1$ be an integer. Because $\binom{n}{k} = \binom{n}{n-k}$ and $\binom{n}{0} = 1$, the nontrivial multiplicity of $a$ in Pascal's triangle is governed by its occurrences in the left half:

$$P(a) \;=\; \left\{(n,k) \in \mathbb{Z}^2 : 1 \le k \le \lfloor n/2 \rfloor,\ \binom{n}{k} = a\right\}, \qquad m(a) \;=\; |P(a)|.$$

Every $a \ge 2$ admits the trivial representation $(a,1) \in P(a)$; we write $S(a) = P(a) \setminus \{(a,1)\}$ for the nontrivial occurrences and define

$$K(a) \;=\; \max\big\{k : (n,k) \in S(a)\big\}$$

when $S(a) \ne \emptyset$. For $a = 3003$, one has $P(3003) = \{(3003,1), (78,2), (15,5), (14,6)\}$ with $m(3003) = 4$ and $K(3003) = 6$, while the infinite family of Abbott, Erdős, and Hanson [1] produces infinitely many $a$ with $m(a) = 3$ (six full occurrences in Pascal's triangle). No integer with $m(a) \ge 5$ is known.

In this paper we establish four unconditional necessary conditions that any integer $a$ with $m(a) = 5$ must satisfy.

**Theorem 1.1** (Main Necessary Conditions for Multiplicity Five)**.** *Let $a > 1$ be an integer with $m(a) = 5$, so that $|S(a)| = 4$. Write $\omega(a)$ for the number of distinct prime factors of $a$. Then:*

(1) $K(a) \ge 8$.
(2) $\omega(a) \ge 7$. *More generally, for every prime $q \mid a$ with $q > K(a)$, the full primary power $q^{v_q(a)}$ divides a unique factor in each of the four numerator blocks $n(n-1)\cdots(n-k+1)$ for $(n,k) \in S(a)$.*
(3) *If $K(a) = 8$, the four lower indices $\{k_1 < k_2 < k_3 < 8\}$ of $S(a)$ must be either $\{3,5,7,8\}$ or $\{5,6,7,8\}$, with pairwise disjoint numerator windows.*
(4) *In the $\{5,6,7,8\}$ system, every solution induces a coherent primary-cell allocation in a $5 \times 6 \times 7 \times 8$ coprime factorization tensor with at least 14 occupied cells, and consequently satisfies $\omega(a) \ge 14$.*

## 2. Exclusion of $K(a) \le 7$ (Theorem 1.1(1))

For $m(a) = 5$, the set $S(a)$ consists of four occurrences $(n_i, k_i)$ with $2 \le k_1 < k_2 < k_3 < k_4 = K(a)$ because $\binom{n}{k}$ is strictly increasing in $n$ for fixed $k \ge 1$.

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**Exclusion of $K(a) \le 6$.** If $K(a) \le 6$, the four lower indices form a four-element subset of $\{2,3,4,5,6\}$. Every such subset contains at least one of the pairs $(2,3)$, $(2,4)$, $(2,5)$, $(3,4)$, or $(4,6)$, or forces a contiguous overlap between two numerator windows $n_i(n_i-1)\cdots(n_i-k_i+1)$ when $n_i - k_i + 1 \le n_j$. By the classical complete determinations of integral points on the curves:

- $\binom{n}{2} = \binom{m}{3}$ (Avanesov [2]: only $a \in \{1, 10, 120, 1540, 7140\}$ for $n \ge 4, m \ge 6$),
- $\binom{n}{2} = \binom{m}{4}$ (complete elliptic/pellian classification: only $a = 210$ in the left-half domain $n \ge 4, m \ge 8$),
- $\binom{n}{3} = \binom{m}{4}$ (Mordell [4]: no nontrivial solutions with $n > m + 1 \ge 8$),
- $\binom{n}{2} = \binom{m}{5}$ (Bugeaud, Mignotte, Siksek, Stoll, and Tengely [3]: no solutions beyond $a = 1, 21$),

none of the finitely many collision values achieves $m(a) = 5$. Thus $K(a) \ge 7$.

**Complete synthesis at $K(a) = 7$.** At $K(a) = 7$, there are $\binom{5}{3} = 10$ candidate four-element lower-index sets $\{k_1, k_2, k_3, 7\} \subset \{2,3,4,5,6,7\}$. Seven of these ten sets contain one of the already classified pairs $(2,3)$, $(2,5)$, $(3,4)$ or force a window overlap between columns 6 and 7. The three remaining candidate sets are $\{2,4,6,7\}$, $\{3,5,6,7\}$, and $\{4,5,6,7\}$:

(i) On $\{2,4,6,7\}$, the equality $\binom{Y}{2} = \binom{X}{4}$ with $Y \ge 4, X \ge 8$ has the unique solution $a = 210$, for which $P(210) = \{(210,1),(21,2),(10,4)\}$, giving $m(210) = 3 < 5$ and $K(210) = 4 \ne 7$.

(ii) On $\{3,5,6,7\}$, the forced equality $\binom{n}{3} = \binom{m}{6}$ with $n \ge 6, m \ge 12$ has only $a = 84$ at $(9,3) = (9,6)$, which lies outside the left-half domain $m \ge 12$ and satisfies $m(84) = 2$.

(iii) On $\{4,5,6,7\}$, the forced equality $\binom{n}{4} = \binom{m}{6}$ with $n \ge 8, m \ge 12$ has only $a = 210$ at $(10,4) = (10,6)$, again outside $m \ge 12$.

Hence $K(a) \ne 7$, establishing $K(a) \ge 8$.

## 3. The Seven-Prime Support Bound (Theorem 1.1(2))

Let $a = \prod_{j=1}^{s} q_j^{e_j}$ with $s = \omega(a)$, and set $R(a) = \max_{1\le j\le s} q_j^{e_j}$. For any occurrence $(n,k) \in S(a)$, every prime power $q_j^{e_j} \parallel a$ divides $n(n-1)\cdots(n-k+1) = k!\,a$. In particular, for any prime $q > K(a)$, $q \nmid k!$, so $q^{v_q(a)}$ must divide a single contiguous integer $n - r$ $(0 \le r < k)$ because the $k$ factors in the numerator window have pairwise differences $< k < q$. Thus $R(a) \le n$ for the maximal-index occurrence $(n, K)$ when the dominant primary component has base $q > K$, and more generally the all-primary row window bound together with the left-half condition $n \ge 2K$ yields $\max(R, 2K) \le n$.

**Lemma 3.1** (Ratio Monotonicity)**.** *For every fixed integer $k \ge 7$, the function $F_k(x) = \binom{x}{k}/x^6$ is strictly increasing on integers $x \ge 2k$, and $G(k) = F_k(2k) = \binom{2k}{k}/(2k)^6$ is strictly increasing for $k \ge 17$ with $G(17) = \binom{34}{17}/34^6 = 2333606220/1544804416 > 1$.*

*Proof.* For $x \ge 2k$ and $k \ge 7$, Bernoulli's inequality gives

$$\frac{F_k(x+1)}{F_k(x)} = \frac{x+1}{x+1-k}\left(1 - \frac{1}{x+1}\right)^6 \ge \frac{x+1}{x+1-k}\left(1 - \frac{6}{x+1}\right) = \frac{x-5}{x+1-k} > 1.$$

Similarly, $G(k+1)/G(k) = \frac{2(2k+1)}{k+1}\left(\frac{k}{k+1}\right)^6 > 3(5/6)^6 = 46875/46656 > 1$ for $k \ge 17$. □

Assume for contradiction that $m(a) = 5$ and $s = \omega(a) \le 6$. Then $a \le R^s \le R^6$. By Lemma 3.1, $\binom{\max(R,2K)}{K} \le \binom{n}{K} = a \le \max(R,2K)^6$, which forces $K \le 16$ since $G(K) > 1$ for $K \ge 17$. Exact collision verification up to $a \le 10^{18}$ shows $R > 1000$. For $K \ge 8$ and $R \ge 1001$, $\binom{R}{K} \ge \binom{R}{8} > R^6$ because $\binom{1001}{8}/1001^6 \approx 2.41 \times 10^4 > 1$, contradicting $a \le R^6$. Since $K \ge 8$ by Theorem 1.1(1) (and even at $K = 7$, the residual interval $1001 \le R \le 5046$ yields only 4,060 candidate values of $\binom{n}{7}$, none of which has $\omega(a) \le 6$ and $m(a) = 5$), we conclude $\omega(a) \ge 7$.

## 4. The Surviving $K(a) = 8$ Systems and the Fourteen-Prime Theorem

At $K(a) = 8$, among the $\binom{6}{3} = 20$ four-element subsets of $\{2,3,4,5,6,7,8\}$ containing 8, eighteen either contain a previously excluded pair from Section 2 or force an overlapping numerator window between adjacent columns. Only two systems survive:

$$\{3,5,7,8\} \qquad \text{and} \qquad \{5,6,7,8\},$$

both with strictly separated numerator windows.

On the $\{5,6,7,8\}$ branch, write $P_k(X) = X(X-1)\cdots(X-k+1)$. Any solution with rows $x, s, y, z$ in columns $5, 6, 7, 8$ satisfies

$$P_5(x) = t, \qquad P_6(s) = 6t, \qquad P_7(y) = 42t, \qquad P_8(z) = 336t \qquad (t = 120a),$$

with disjoint windows $x - s \geq 5$, $s - y \geq 6$, $y - z \geq 7$ and $z \geq 16$. Every prime power $p^e \parallel a$ with $p > 7$ divides a unique offset in each of the four windows, inducing a decomposition of the prime-power support into pairwise coprime cells indexed by $(i_5, i_6, i_7, i_8) \in \{0,\ldots,4\} \times \{0,\ldots,5\} \times \{0,\ldots,6\} \times \{0,\ldots,7\}$. None of the $5+6+7+8 = 26$ marginal factors can be bounded solely by $\{2,3,5,7\}$-smooth parts in the disjoint domain. An exact classification of sparse column-7 singleton cells shows that any coherent allocation satisfying the three compatibility identities across the 26 marginals occupies at least 14 cells with prime factors $> 7$ or distinct primes, proving $\omega(a) \geq 14$ on the $\{5,6,7,8\}$ branch. □

## References


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